International Journal of Digital Earth

ISSN: 1753-8947 (Print) 1753-8955 (Online) Journal homepage: www.tandfonline.com/journals/tjde20

Gully-ERFNet: a novel lightweight deep learning model for extracting erosion gullies in the black soil region of Northeast China

Qingyao Li, Jiuchun Yang, Jiaqi Wang, Zhi Li, Jianwei Fan, Liwei Ke & Xue Wang

To cite this article: Qingyao Li, Jiuchun Yang, Jiaqi Wang, Zhi Li, Jianwei Fan, Liwei Ke & Xue Wang (2025) Gully-ERFNet: a novel lightweight deep learning model for extracting erosion gullies in the black soil region of Northeast China, International Journal of Digital Earth, 18:1, 2494074, DOI: 10.1080/17538947.2025.2494074

To link to this article: https://doi.org/10.1080/17538947.2025.2494074

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 21 Apr 2025.

Submit your article to this journal

Article views: 1409

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tjde20

INTERNATIONAL JOURNAL OF DIGITAL EARTH
2025, VOL. 18, NO. 1, 2494074
https://doi.org/10.1080/17538947.2025.2494074

OPEN ACCESS Check for updates

Gully-ERFNet: a novel lightweight deep learning model for extracting erosion gullies in the black soil region of Northeast China

Qingyao Li[a], Jiuchun Yang [a], Jiaqi Wang[a,b], Zhi Li[a,b], Jianwei Fan[a,c], Liwei Ke[a] and Xue Wang[a]

[a]State Key Laboratory of Black Soils Conservation and Utilization, Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun, People's Republic of China; [b]University of Chinese Academy of Sciences, Beijing, People's Republic of China; [c]Jilin Agricultural University, Changchun, People's Republic of China

ABSTRACT
Gully erosion is a critical environmental issue in the black soil region of Northeast China, causing severe soil degradation and posing significant threat to food security. Accurate gully extraction is essential for implementing effective control measures to mitigate environmental and agricultural impacts. Traditional machine learning methods rely on high-resolution digital elevation models (DEMs) but are hindered by data acquisition challenges and low automation levels. To resolve this issue, we introduce the Gully-ERFNet model, designed to extract gullies without high-resolution DEMs. Based on the deep learning ERFNet architecture, the Gully-ERFNet model integrates the Squeeze-and-Excitation attention mechanism to expand the global receptive field and the Pyramid Pooling Module to improve the model's emphasis on gully features. We employed the Adaptive Correction Loss function to reduce the impact of incorrect labels on model performance. The model's performance was evaluated using GF-2 satellite remote-sensing images of the Hailun Basin. The model achieved an F1-score of 81.50%, recall rate of 76.24%, and accuracy of 87.54% in gully extraction, outperforming existing gully extraction algorithms, such as Random Forest, U-Net, and DeeplabV3+. This model enables efficient gully extraction using only remote sensing imagery, addresses difficulties in data collection, and provides support for preventing and controlling gully erosion.

ARTICLE HISTORY
Received 6 November 2024
Accepted 5 April 2025

KEYWORDS
Gully extraction; digital elevation model; deep learning; gully-ERFNet; attention mechanism

1. Introduction

Erosion gullies are channels formed by surface water flow, typically caused by heavy rainfall or snowmelt, which erodes the soil and forms channels (Sidorchuk 1999; Tang et al. 2021; Thwaites et al. 2021). These gullies transport large amounts of soil as they develop, resulting in significant loss of land (Dagar and Singh 2018; Xu et al. 2010). The black soil region of Northeast China, one of the largest black soil regions in the world, is a crucial region for grain production in

CONTACT Jiuchun Yang yangjiuchun@iga.ac.cn State Key Laboratory of Black Soils Conservation and Utilization, Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun, Jilin 130102, People's Republic of China

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

China, contributing to a quarter of the total grain production (Wen et al. 2021). However, the unique climatic conditions of the region are characterized by cold and dry winters, followed by summers with intense rainfall (Tang et al. 2022; Tang et al. 2023), leading to extensive surface erosion in Northeast China. Unsustainable land use and excessive cultivation further exacerbate surface and gully erosion (Liu et al. 2005), significantly impacting local grain production and regional ecological security (Ma et al. 2024; Liu et al. 2012; Wang et al. 2023). Statistics show that Northeast China's black soil region contains approximately 667,000 permanent gullies over 2 m in length, 75% of which are located on farmland. Exacerbated slope erosion and soil degradation have resulted in land becoming unsuitable for cultivation and caused an annual loss of approximately 36.2 billion kilograms of grain (Gao et al. 2024). Therefore, the prevention and control of gully erosion in this region is urgently needed, making the identification of its spatial distribution a necessary prerequisite for the selection and implementation of effective gully prevention and control measures.

Currently, the most widely used methods for gully identification include field surveys (Castillo et al. 2012; De Santisteban, Casalí, and López 2005), visual interpretation (Wang et al. 2016), and machine-learning-based methods (Wang et al. 2020). Traditional field surveys, although mostly highly accurate, are time-consuming and labor-intensive. Visual interpretation is more efficient than field surveys using remote sensing imagery but is still constrained by a high level of subjectivity (Gao 2009).

Building on the limitations of traditional approaches, machine learning methods have been introduced to automate gully identification, including pixel-based image analysis (PBIA) (Neupane, Horanont, and Aryal 2021) and object-based image analysis (OBIA) (Mollick, Azam, and Karim 2023). PBIA methods analyze each pixel in an image and classify and identify gullies based on various pixel-level features (Hossain and Chen 2019). However, PBIA methods are limited by their reliance on individual pixel information, which neglects spatial relationships and contextual information, making them prone to noise and misclassification (Blaschke et al. 2014). In contrast, OBIA segments an image into larger objects or regions by considering the spatial relationships and contextual information among the pixels (Huang, Xiong, et al. 2023). Although OBIA can improve accuracy by incorporating topography and texture (Garosi et al. 2018; Karami et al. 2015; Shahabi et al. 2019), it faces challenges in parameter optimization. The sensitivity of OBIA to parameter selection makes it difficult to define the optimal settings for gullies, which often exhibit diverse shapes and sizes and can be visually similar to the surrounding terrain.

Given these challenges of traditional and machine learning methods, there is a growing need for more flexible and automated techniques. Deep learning models, with their ability to automatically learn features and adapt to complex data patterns, offer promising solutions to these challenges. Specifically, Convolutional Neural Networks (CNN) (Gu et al. 2018) and recurrent neural networks (RNNs) (Grossberg 2013) have enabled computers to recognize deeper features and perform well in gully extraction tasks (Liu et al. 2022). An increasing number of remote sensing scientists have begun to apply deep learning techniques in this field. For example, U-Net has been successfully applied to extract gullies in the Eastern Plains of Russia (Gafurov and Yermolayev 2020), while an enhanced DeeplabV3+ network has shown promising results on China's Loess Plateau (Chen et al. 2024). Additionally (Zhu et al. 2024), improved the LinkNet model by better integrating contextual information around gullies in the northeastern region, enhancing extraction accuracy. However, despite these advancements, existing deep-learning approaches still face significant limitations in accurately extracting gullies. Subtle differences in texture, spectral reflectance, and elevation between gullies and their surrounding terrain result in low contrast, hindering the ability of standard convolutional layers to reliably distinguish gully boundaries. The considerable diversity in gully size, shape, and complexity coupled with their relatively small areal extent introduces a class imbalance problem. These factors contribute to models struggling with the accurate identification of small or narrow gullies and the frequent misclassification of other linear features, such as

roads, ditches, and field boundaries (Chen et al. 2024; Fathololoumi et al. 2025; Liu et al. 2022; Zhang et al. 2024). Although some studies have incorporated Digital Elevation Model (DEM) data to enhance accuracy (Huang, Xiong, et al. 2023; Shen et al. 2024), the acquisition of high-resolution DEM data presents significant cost and accessibility barriers, limiting its widespread use in large-scale applications (Li et al. 2021). This underscores the need for continued advancements to ensure that deep learning models can accurately identify gullies in diverse and complex landscapes, ideally without relying on high-resolution DEM data.

To address this requirement, this study introduced Gully-ERFNet, a novel deep-learning model built on an efficient ERFNet architecture (Romera et al. 2018). The lightweight design and proven ability of ERFNet to capture fine details and small objects (Zhang et al. 2021) make it suitable for delineating the complex morphology of erosional gullies, especially in resource-constrained environments. Furthermore, the modular design of ERFNet facilitates the integration of enhancements. Building on this foundation, Gully-ERFNet incorporates attention mechanisms and a pyramid pooling module to enhance feature extraction and address the challenges of scale and subtle feature contrast. This approach offers a potentially practical and cost-effective solution for large-scale gully mapping using readily available imagery. The main contributions of this study are as follows.

- A novel neural network, Gully-ERFNet, was constructed for gully erosio
- n identification based on the ERFNet model. The ability of the model to capture critical information about erosion gullies was enhanced by expanding the receptive field through the introduction of the Pyramid Pooling Module (PPM) and incorporating the Squeeze-and-Excitation (SE) attention mechanism.
- We designed the ACL function to suppress imbalances and noise issues in gully samples.
- The proposed Gully-ERFNet model was applied to gully extraction in the black soil region of Northeast China, and its effectiveness was validated through comparisons with the results from RF, U-Net, and DeeplabV3+, as well as through module ablation experiments.

2. Materials and methods

2.1. Study areas

In this research, Hailun City, Heilongjiang Province, China, was selected as the study area for gully extraction as shown in Figure 1. This region lies at the heart of a typical black soil zone covered by deep, dark-colored, and organic-rich topsoil with an average thickness of over 70 cm. The region experiences a temperate continental monsoon climate, featuring cold winters with heavy snowfall and hot, rainy summers, during which the annual rainfall ranges between 500 and 600 mm (Zhang et al. 2022). Additionally, terrain with slight slopes and large catchment areas contributed significantly to water erosion from rainwater runoff. Agricultural land is the major land use type in the study area (63.3%), and long-term unsustainable farming practices have diminished the capacity of sloping land to intercept runoff. This decline exacerbates soil erosion and promotes gully formation, ultimately threatening the agricultural productivity in the region.

To ensure that the model could accurately identify gullies under diverse geographical conditions within the black soil region and demonstrate both robustness and extensibility, two subareas, Hailun Basin and Zake-Keyin Basin in Hailun City, were selected to test the deep learning model (Figure 1). Hailun Basin (127°04′–127°32′E, 47°47′–47°58′N) spans 165.11 km^2 with an elevation ranging from 138 to 249 m. In contrast, Zake-Keyin Basin (126°84′–127°07′E, 47°22′–47°37′N) covers 196.04 km^2 with elevations between 177 and 314 m. Both subareas contain a variety of gully types, including rills, shallow gullies, cut gullies, and ravines, with different terrain characteristics. The Hailun Basin has larger cultivation plot sizes with relatively small and homogenous undulations. Zake-KeyinBasin exhibits notable terrain heterogeneity, with varying slopes, aspects, and plot sizes, owing to its proximity to mountainous areas.

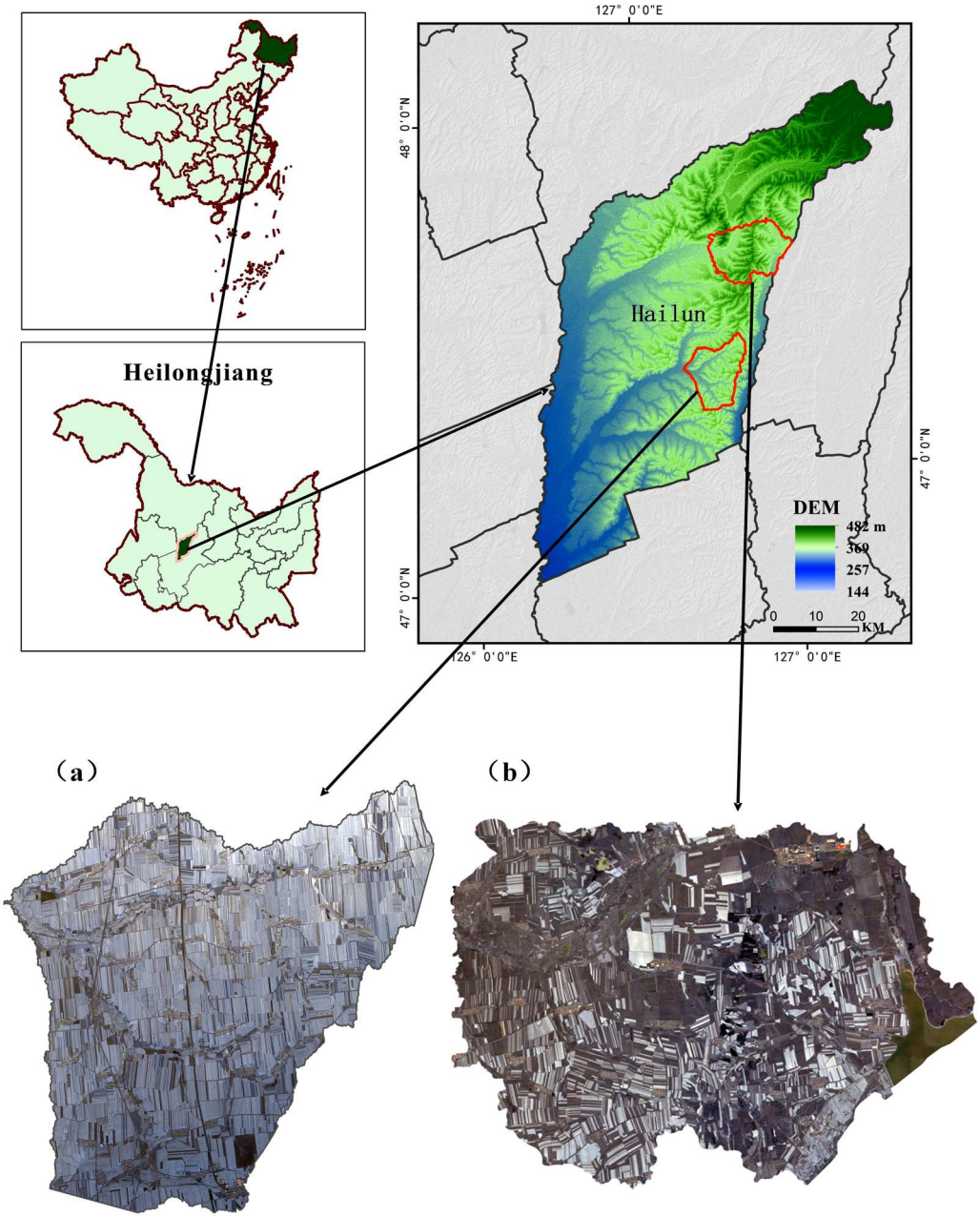

Figure 1. Location of the study areas: (a) Hailun basin. (b) Zake-Keyin basin.

2.2. Data source

In this study, GF-2 remote sensing images were selected for gully extraction. These images consist of four multispectral bands with a spatial resolution of 3.2 m, alongside one panchromatic band with a resolution of 0.8 m (Chen et al. 2022; Li, Wang, and Jiang 2020). The different spectral bands capture various characteristics of ground objects, and when fused, they provide a composite image with a resolution of 0.8 m (Ghassemian 2016). This high-resolution imagery enables a clear

distinction between erosion gullies and their surrounding surface features, significantly enhancing the classification accuracy.

Since vegetation during the growing season can obscure gullies, images captured during bare land periods are more suitable for accurate extraction. Therefore, the GF-2 images captured in November, after the autumn harvest, were chosen as the data source for this study. At this time, vegetation cover was minimal, enhancing the visibility of the eroded gullies and ensuring that the ground was free from snow cover. GF-2 images for the Hailun and Zake-Keyin basins were acquired on November 4, 2022, and November 4, 2023, respectively. These images, taken at different times, enable the model to account for temporal and environmental changes, providing robust support for scalability and adaptability.

In addition, to evaluate the impact of Digital Elevation Model (DEM) data on gully erosion extraction, we used DEM data with a resolution of 12.5 m for comparative experiments. DEM data were obtained from the Alaska Satellite Facility (ASF) website. Incorporating DEM data into comparative experiments enables a more comprehensive analysis and facilitates a deeper understanding of how terrain features affect gully extraction performance.

2.3. Erosion gully extraction method

To efficiently extract small targets and boundary details, this study adopted the ERFNet model as the foundation for identifying erosional gullies. Given the unique characteristics of gullies in Northeast China's black soil region, we enhanced the ERFNet architecture to create a new deep-learning semantic segmentation model called Gully-ERFNet. This model integrates the SE attention mechanism, which expands the global receptive field, with the PPM, thereby enhancing the model's focus on gully features. This strategic combination aims to significantly improve the overall gully extraction performance.

The process of extracting erosion gullies using the Gully-ERFNet model is illustrated in Figure 2. Initially, erosion gully data underwent preprocessing, followed by PCA (Maćkiewicz and Ratajczak 1993) and multi-scale segmentation (Kotaridis and Lazaridou 2021) to obtain preliminary classification results. Subsequently, images were cropped into 512 × 512 pixel blocks along with their corresponding labels. The dataset was further refined using data augmentation techniques including flipping, rotation, translation, and contrast enhancement, before being divided into training, validation, and test sets. During the training phase, image blocks from the training set were inputted into the Gully-ERFNet model. Throughout this stage, model performance was meticulously evaluated using the validation set, ensuring loss convergence and preventing overfitting. Once trained, the model was applied to the test dataset to produce a probability map illustrating gully erosion. The extracted data underwent post-processing, utilizing remove_small_objects and remove_small_holes morphological operations to enhance the gully extraction accuracy. Finally, the model's capabilities were evaluated using metrics such as accuracy, recall, and F1 scores, which affirmed its robustness and generalization in real-world scenarios.

In the following subsections, each step of the workflow is explored in detail to gain a comprehensive understanding of the methods and techniques used at each stage.

2.3.1. Data preprocessing and sample generation

Owing to the limited availability of open-source gully erosion data samples, most dataset generation relies on the manual interpretation of remote sensing images. Moreover, this process is highly labor-intensive and susceptible to human-induced variability, as different researchers may interpret gully features differently, leading to subjective results in gully annotations (Chowdhuri et al. 2021; Phinzi, Holb, and Szabó 2021). To mitigate the costs associated with manual annotations, we used the GF2 remote sensing imagery for preprocessing and sample generation, as illustrated in Figure 2(a).

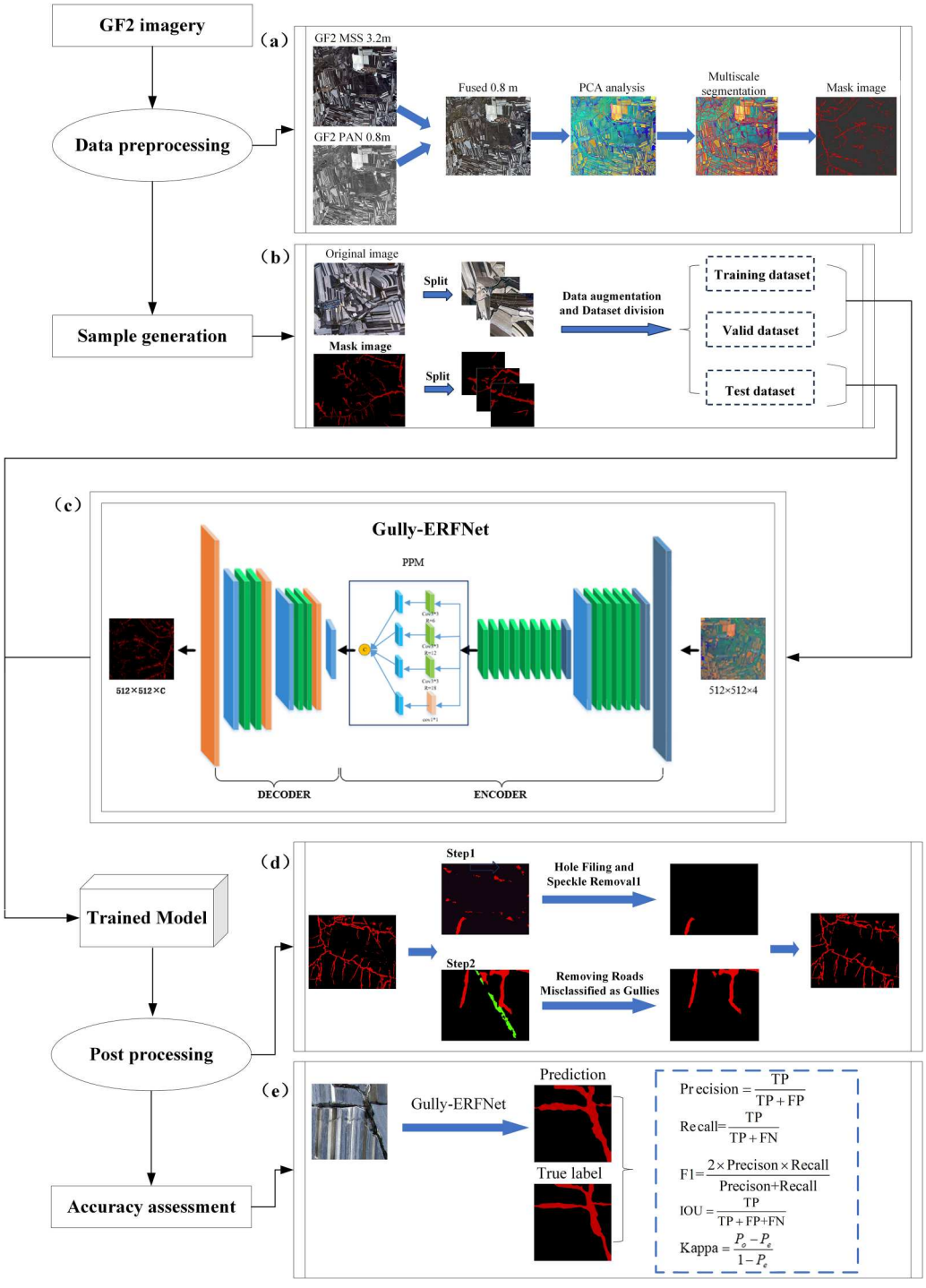

Figure 2. Workflow for erosion gully extraction. The entire process includes (a) Data preprocessing, (b) Sample generation for training the Gully-ERFNet model, (c) applying the trained model to extract erosion gullies in the Hailun and Zake-Keyin basins, followed by (d) Post-processing and (e) Accuracy assessment to evaluate the model's performance.

Initially, the GF2 images were fused to generate four-band multispectral images with a resolution of 0.8 m (Wang, Ma, and Zhang 2023). Subsequently, the PCA function in the ENVI software was used to reduce data dimensionality while retaining the main information from the images and key features of erosion gullies, facilitating subsequent image segmentation (Estornell et al. 2013). Subsequently, the PIE software, an OBIA technique, was utilized to perform multiscale image segmentation. This method divides images into homogeneous segments and selects gully regions to generate preliminary segmented gully masks (Shivhare and Gupta 2015). To further ensure the accuracy of the results were cross-validated with data from the Third National Land Survey and higher-resolution aerial images to ensure the reliability of the ground truth dataset. This classification method partially replaces manual annotation and effectively analyzes remote-sensing images, thereby providing solid technical support for future research.

The sample generation process is illustrated in Figure 2(b). In the generated gully mask image, the gully regions were labeled 1, and non-gully regions were labeled 0. These mask images, along with their corresponding original images, were trimmed into 512×512 pixel blocks to form image-label pairs. To prevent overfitting, data augmentation techniques such as flipping, rotation, translation, and contrast enhancement were applied, expanding the dataset to 6,400 pairs. Finally, these augmented image-label pairs were divided into training, validation, and testing sets at an 8:1:1 ratio for model training.

2.3.2. Gully-ERFNet model construction

The traditional semantic segmentation network models have been designed to extract linear objects from images; however, it is difficult to distinguish gullies from other complete linear objects such as roads. This similarity makes it difficult for mainstream semantic segmentation models to accurately delineate gully boundaries. To address this issue, we propose a novel neural network, Gully-ERFNet, which is specifically designed for gully erosion identification. This model was built upon the ERFNet semantic segmentation framework and incorporated the PPM (Yu, Yang, and Chen 2018) to expand the receptive field by pooling features at multiple scales, which helped capture contextual information from various regions of the image and improve image perception. Furthermore, SE (Hu, Shen, and Sun 2018) was integrated to better capture gully characteristics and minimize interference from redundant information. The Gully-ERFNet model framework is illustrated in Figure 3.

Gully-ERFNet adopted the encoder-decoder architecture of the original ERFNet model. The encoder consisted of 3 downsampling modules and 13 Non-bottleneck-1D modules, while the decoder comprised 3 upsampling modules and 4 Non-bottleneck-1D modules. As illustrated in Figure 4(a), the downsampling module reduced the spatial resolution of feature maps by combining 3×3 convolutions with a stride of 2 and max pooling operations. The upsampling module shown in Figure 4(b) employed 3×3 convolutions with a stride of 2, followed by batch normalization and ReLU activation to restore feature maps to higher resolutions. The Non-bottleneck-1D module depicted in Figure 4(c) served as the core component of ERFNet, designed to extract richer features while maintaining computational efficiency. This module decomposed standard 3×3 convolutions into sequential 1×3 and 3×1 convolutions, and incorporated residual connections that added input feature maps to output feature maps, thereby facilitating gradient flow and accelerating network training.

To further enhance model performance, we integrated SE attention mechanism modules after the convolutional layers in both encoder and decoder. This design is crucial for enhancing the model's ability to discriminate between gullies and visually similar features (e.g. roads and small streams), and to emphasize subtle gully characteristics that might be obscured by noise or variations in illumination. Unlike many other linear features, gullies often exhibit subtle channel-specific variations and texture. The SE module addresses this by adaptively recalibrating the channel-wise feature responses. It can explicitly model the interdependencies between feature channels, potentially allowing the network to selectively emphasize channels that are most informative for gully detection

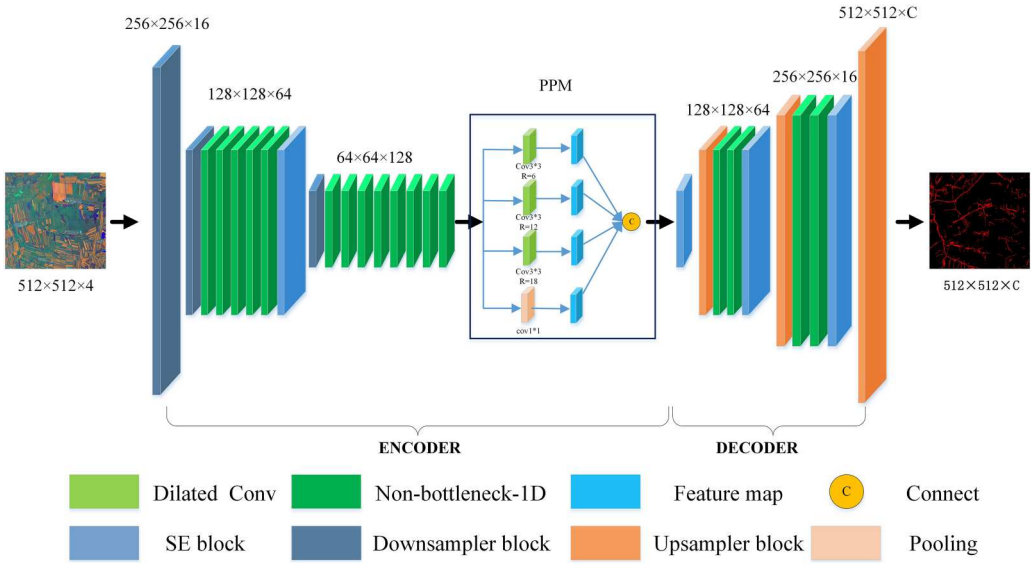

Figure 3. Gully-ERFNet: Enhancing erosion gully detection through advanced SE Blocks and Pyramid Pooling Modules in a Deep-Learning framework.

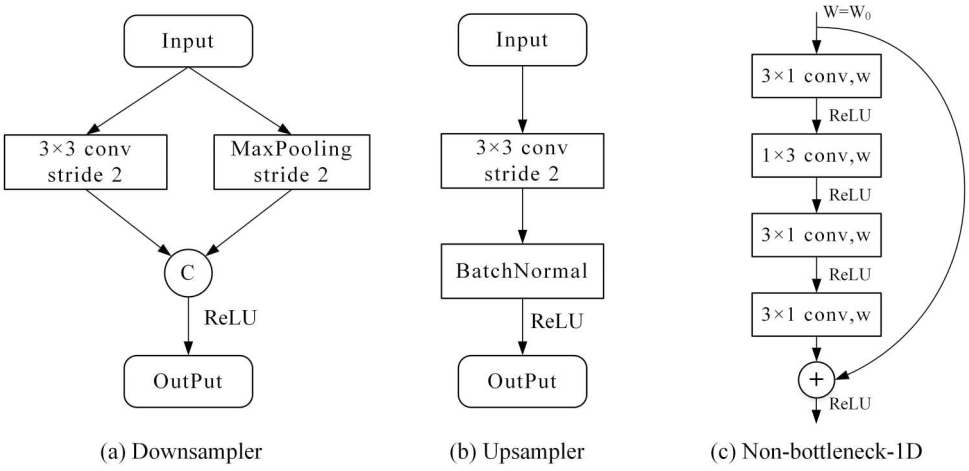

Figure 4. Core components of the ERFNet architecture: (a) Downsampler block, (b) Upsampler block, (c)Non-bottleneck-1D block.

while suppressing less relevant or noisy channels. This targeted emphasis may improve the representation of gully specific characteristics, potentially leading to a more precise delineation of boundaries. The integration of this mechanism enabled the model to focus better on critical gully features, thereby enhancing its overall performance.

As shown in Figure 5, the SE module operates in three main steps: Squeeze (F_{sq}), Excitation (F_{ex}), and Scaling (F_{scale}). Initially, the Squeeze step employs global average pooling to condense the spatial information of the original input feature map, with dimensions $H \times W \times C$, resulting in a feature map of size $1 \times 1 \times C$. This vector then enters the excitation step, where it is refined using two fully connected layers. The first layer condenses the information using ReLU activation and identifies salient features. The second layer used a sigmoid activation function to restore the original channel dimensions to $1 \times 1 \times C$, yielding a set of weights representing the significance

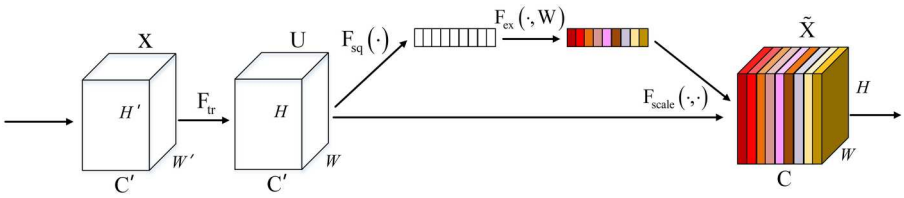

Figure 5. The architecture of SE attention mechanism in Gully-ERFNet model.

of each channel. In the Scaling step, the resulting weight vector is multiplied by the original feature map with dimensions H × W × C to adjust for the influence of each channel. This process amplifies the presence of significant features while attenuating those that are less critical.

In the ERFNet encoder, we added the SE module after the last two sets of convolution operations to adaptively adjust the channel weights of the features extracted through the convolutional layers, thereby enhancing the important features extracted by the encoder. In the decoder, the SE module was added after the convolutional layers to adaptively adjust the channel weights, thereby enhancing the representation of key features during the feature recovery process. This means that during the step-by-step decoding and resolution recovery processes, the SE module adaptively adjusts the feature channels, helping the model better preserve key information.

In addition to the SE mechanism, to address the challenge of varying gully sizes and shapes and to capture multiscale contextual information, we incorporated the PPM between the encoder and decoder (Figure 3). Gullies range from small, narrow features to large, complex networks. By pooling features at multiple scales, PPM provides a global context that is crucial for distinguishing gullies from other linear features and accurately capturing their full extent. The PPM generates multiple representations of the encoder output by applying pooling operations at different scales (Zhao et al. 2017). These multi-scale feature maps are then concatenated and processed to create a rich, context-aware representation. This approach mitigates the loss of spatial information inherent in standard pooling operations and improves the model's ability to capture both local details and global context, leading to more precise gully delineation.

The combination of the SE and PPM modules within the Gully-ERFNet provided a synergistic effect. The PPM broadens the receptive field of the model, enabling it to capture multiscale contextual information that is crucial for understanding the overall landscape and identifying gully networks. The SE modules, on the other hand, refine the feature representations at both the encoding and decoding stages, ensuring that the model focuses on the most discriminative features for gully identification. This integrated approach enhances both the accuracy and robustness of gully extraction, reduces computational costs compared to larger, more complex models, and improves generalization performance across diverse landscapes.

2.3.3. Design of ACL function

During the training step in the Gully-ERFNet model, noise can easily be involved owing to inaccuracies in sample selection for object-oriented segmentation. In addition, the small proportion of eroded gully areas in the images created an imbalance between the gully and non-gully samples. To address these issues, we have designed an Adaptive Correction Loss (ACL) function. This loss function is dynamically adjusted based on the convergence of the model and the data distribution during training. Thus, the ACL function helps the model adapt to different scenarios, mitigating the effects of noise and sample imbalances. The loss function is given by Equation (1).

$$ACL = \lambda \cdot Loss_{Dice} + (1 - \lambda) \cdot Loss_{Focal} \tag{1}$$

The ACL function is composed of Dice and the Focal loss functions. Where λ adjusts the balance between Dice and Focal losses during training. $Loss_{Dice}$ (Xu et al. 2023) and $Loss_{Focal}$ (Lin et al. 2017) denote the Dice loss function and the Focal loss function, respectively.

The Dice loss function emphasizes the predictive accuracy of the model at the boundary positions, making it particularly effective for handling areas with ambiguous or irregular shapes (Jadon 2020). Because gullies often have complex shapes and irregular boundaries, incorporating the Dice loss function in gully extraction can improve the model's prediction of gully boundaries. The Dice loss function is given by Equation (2):

$$Loss_{Dice} = 1 - \frac{2\sum_{i=1}^{N} pi*gi}{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} g_i^2} \tag{2}$$

where N is the total number of pixels, and pi and gi represent the predicted value and true label value of pixel i, respectively. Lower Dice Loss values indicate greater similarity.

In gully extraction, there is often a significant disparity in the number of pixels between gully and non-gully areas, leading to an imbalance between the positive and negative samples. By employing the Focal Loss function, the model can dynamically adjust the loss function based on class weights, allowing it to focus more on predicting gully areas, thereby improving the precision of gully recognition. The formula is shown in Equation (3):

$$Loss_{Focal} = -\alpha_t(1 - p_t)^{\gamma} \log(p_t) \tag{3}$$

Where αt represents the initial weight of the loss function in the class, γ represents the weight correction coefficient of the positive and negative samples, which is used to reduce the weight of the positive. When γ equals 1 Focal loss degrades to a cross-entropy loss function.

2.3.4. Post-processing

Although deep learning models demonstrate high performance in the extraction of erosion gullies, noise, and other factors often introduce inaccuracies. These inaccuracies led to the presence of small artifacts and holes in the results. Furthermore, the textural similarities between roads and erosion gullies frequently result in roads being misclassified as gullies. To overcome this problem, essential post-processing operations are implemented in two main steps. The first step involved addressing small patches and holes. Analysis of the training data revealed that most erosion gullies exceeded 1000 pixels (approximately 140 m^2). Therefore, the remove_small_objects and remove_small_holes functions from the scikit-image library in Python (Singh and Dhir 2019) were utilized to perform speckle removal and hole filling, with the parameters set to 1000 pixels.

The second step focused on correcting misclassified areas such as roads and residential ditches. For road sections mistakenly classified as gullies, we utilized basic geographic vector data at a 1:25,000 scale from the National Basic Geographic Information Center. A 20 m buffer was applied to the road data, converting road lines into polygons to exclude them from the gully classification. Similarly, for misclassified residential areas and field ditches, we used urban vector data and applied the erase function in ArcGIS 10.3 to correct these errors.

2.3.5. Model evaluation metric

To effectively evaluate the model performance in gully extraction, we employed common semantic segmentation metrics: Precision, Recall, Intersection over Union (IoU), F1-score, and the Kappa coefficient. The precision reflects the model's false positive rate of the model, with higher values indicating fewer non-gullies misclassified as gullies. The recall measures the proportion of true gullies correctly identified, where a higher recall indicates fewer missed gullies. In gully extraction tasks, IoU specifically represents the proportion of the overlapping area between the predicted and actual gully boundaries relative to their total area. Since gully boundaries are often complex and irregular, IoU effectively evaluates the model's precision in delineating these boundaries. Higher IoU values indicate that the model accurately delineates gully boundaries, enhancing spatial consistency and practical applicability of the extraction results. The F1-score balances Precision and Recall, offering a comprehensive view of the model's detection capability in complex terrains.

Additionally, the F1-score reflects the model's ability to balance false negatives and false positives, which is crucial for the accuracy of gully detection. The Kappa coefficient measures the agreement between model predictions and random classifications, correcting for chance agreement. This reflects the overall accuracy and assesses the robustness of the model against data imbalance and noise. The calculation formulae are provided in Equations (4)–(8).

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{4}$$

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{5}$$

$$IoU = \frac{\mathrm{TP}}{\mathrm{FP} + \mathrm{TP} + \mathrm{FN}} \tag{6}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{7}$$

In the above equation, True Positive (TP) indicates the number of positive samples correctly predicted as positive by the model; True Negative (TN) represents the number of negative samples correctly predicted as negative by the model, indicating the model's ability to accurately identify non-gully areas; False Positive (FP) indicates the number of negative samples incorrectly predicted as positive; and False Negative (FN) indicates the number of positive samples incorrectly predicted as negative. The Kappa coefficient is calculated as follows:

$$Kappa = \frac{P_o - P_e}{1 - P_e}. \tag{8}$$

In the Kappa formula, P_o represents the observed agreement, that is the proportion of the total samples in which the model's predictions match the ground truth, as shown in Equation (9). P_e denotes the expected agreement by chance, calculated based on the marginal probabilities of each class, as shown in Equation (10), where N represents the total number of samples.

$$P_o = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{N}} \tag{9}$$

$$\mathrm{P_e} = \left(\frac{(\mathrm{TP} + \mathrm{FP}) \times (\mathrm{TP} + \mathrm{FN})}{\mathrm{N}^2}\right) + \left(\frac{(\mathrm{FN} + \mathrm{TN}) \times (\mathrm{FP} + \mathrm{TN})}{\mathrm{N}^2}\right) \tag{10}$$

Using these metrics, the performance of deep learning models in extracting erosion gullies can be comprehensively assessed to guide model development and further improve post-processing techniques.

2.4. Experimental design

To validate the effectiveness of the improved gully erosion extraction model, we conducted comparative experiments and ablation studies using samples extracted from GF2 remote-sensing imagery. In the ablation experiments (Zeiler and Fergus 2014), we systematically removed the PPM and SE modules to analyze the performance changes at each stage and determined the contribution of each module. Specifically, four models were compared: the ERFNet model, ERFNet + PPM, ERFNet + SE, and Gully-ERFNet (i.e. ERFNet + PPM + SE), to comprehensively evaluate the contribution of each module. In addition, an ACL function curve is plotted to explore the impact of the loss function on the performance of the model.

In the comparative experiments, the proposed model was compared with existing methods such as RF (Rigatti 2017), U-Net (Ronneberger, Fischer, and Brox 2015), and Deeplab V3+ (Chen et al.

Table 1. Information on the configuration of software and hardware in the experimental environment.

Configuration	Version
System	Windows 10
CPU	Intel(R) Core(TM) i9-12900K
GPU	NVIDIA GTX3070
Memory	64GB 4800MHz
Hard drive	10TB 7200 r/min
CUDA	Cuda 11.0
Frame	Pytorch 1.9

2018) to evaluate performance differences. Moreover, to evaluate the impact of DEM data on the accuracy of gully erosion extraction, this study resampled DEM data with a resolution of 12.5 meters to 0.8 meters using bilinear interpolation, aligning it with the resolution of GF2 imagery. The resampled DEM data were then integrated as an additional input channel with GF2 multispectral imagery. By comparing the performance of the Gully-ERFNet model before and after incorporating DEM data, we quantitatively analyzed the contribution of topographic information to gully extraction results.

To ensure the fairness of the experiments, the experimental environment was standardized and the input image was resampled to a size of 512 × 512 pixels. After multiple experiments, all models converged after 70 iterations; hence, the number of training iterations was set to 100, with four training samples per batch. The optimizer used was Stochastic Gradient Descent (SGD) (Cong and Buratti 2018), with an initial learning rate of 0.005 and the same learning rate decay mechanism applied. All models used the same training, testing, and validation sets, with all other parameters set to their default values. The hardware environments used in the experiments are listed in Table 1.

3. Results

3.1. Module ablation experiment

Ablation experiments were conducted to further evaluate the effectiveness of the PPM and SE modules incorporated in this study. By systematically removing the PPM and SE modules, we analyzed the performance at each stage to assess the contribution of each module. The evaluation utilized the same metrics described in Section 2.3.5, and the results are presented in Table 2.

As shown in Table 2, the introduction of the PPM module led to improvements in all evaluation metrics of the model, including an increase in the Kappa coefficient from 0.763 to 0.782, indicating enhanced classification consistency. The IoU also increased from 62.40% to 65.47%, an increase of 3.07%, highlighting the contribution of the PPM module to better spatial accuracy. When the SE attention mechanism was introduced independently, the precision and F1-score increased to 86.32% and 79.07%, respectively, and the Kappa coefficient improved to 0.788. Although the recall and IoU enhancements were modest, the SE module effectively strengthened the focus of the model on important features, thereby improving the overall classification consistency. Combining both the PPM and SE modules, the Gully-ERFNet model achieved optimal performance across all metrics: a precision of 87.54%, recall of 76.24%, F1-score of 81.50%, IoU of 68.80%, and Kappa of 0.826. These results clearly demonstrate the advantages of integrating PPM with the SE attention mechanism for improved segmentation accuracy and consistency.

Table 2. Ablation comparison experimental results of PPM and SE modules on Gully-ERFNet.

Model	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	Kappa
ERFNet	83.69	71.04	76.84	62.40	0.763
ERFNet + PPM	85.61	73.57	79.13	65.47	0.782
ERFNet + SE	86.32	72.94	79.07	65.38	0.788
Gully-ERFNet	87.54	76.24	81.50	68.80	0.826

3.2. Comparisons with other methods

In the comparative experiments, this study employed the evaluation metrics outlined in Section 2.3.5 to conduct a comprehensive comparison between the extracted erosion gully extraction results of each model and the ground truth values in the test set. This methodology facilitated the quantification and analysis of performance differences among the models and assessed the impact of integrating DEM data on the Gully-ERFNet model's capacity to extract erosion gullies. The evaluation metrics are listed in Table 3.

Based on the evaluation metrics presented in Table 3, the following conclusions can be drawn. In the task of gully erosion extraction, the Random Forest model was found to have the weakest performance, with precision, recall, and Kappa coefficient rates of only 67.68%, 59.81%, and 0.597, respectively, both of which are lower than those of the other models. This suggests that traditional machine learning methods are less effective for complex image segmentation tasks. As a classic semantic segmentation model, U-Net achieves an F1-score of 70.67% and a kappa coefficient of 0.677; however, its IoU value of 54.64% was lower than that of other deep-learning models, indicating significant room for improvement. ERFNet demonstrated better performance than U-Net, with a Precision of 83.69%, an F1-score of 76.84%, and a Kappa coefficient of 0.763, reflecting improved consistency and reliability. Both DeeplabV3+ and Gully-ERFNet exhibit impressive precision, recall, F1-score, IoU, and Kappa coefficient, with Gully-ERFNet exhibiting a precision of 87.54%, representing an improvement of 3.85% over its predecessor. Furthermore, the recall, F1-score, IoU, and Kappa coefficient of Gully-ERFNet were 76.24%, 81.50%, 68.80%, and 0.826, respectively, indicating its superior capability to accurately identify gully areas in the extraction task, thereby highlighting its significance in practical applications.

Table 4 compares the evaluation metrics of ERFNet with and without the inclusion of DEM data. Notably, after fusing the DEM data, the precision and recall for gully erosion extraction experienced negligible variation, with precision decreasing slightly from 87.54% to 86.81% and recall increasing marginally from 76.24% to 76.61%. Similarly, the Kappa coefficient showed a slight decline from 0.826 to 0.821, suggesting that the inclusion of DEM data contributed little to improving the classification consistency in gully erosion extraction.

To demonstrate the advantages of our research model for gully extraction visually, we selected four representative results from a test set. Figure 6 illustrates these findings; the first column in each image group displayed the original remote sensing image, the second column showed the true erosion gully labels (red areas indicate gullies), and the subsequent columns presented the extraction results from various models. The RF model results are relatively coarse and display multiple discontinuities. U-Net and DeepLab V3+ exhibited better continuity, although some areas still had discontinuity issues during extraction. ERFNet identified erosion boundaries more effectively; however, there is still room for improvement in capturing finer details. The Gully-ERFNet model aligned

Table 3. Comparison of the proposed method and state-of-the-art methods using the test set.

Model	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	Kappa
RF	67.68	59.81	63.50	46.52	0.597
U-Net	76.12	65.95	70.67	54.64	0.677
DeeplabV3+	85.21	74.47	79.47	65.95	0.774
ERFNet	83.69	71.04	76.84	62.40	0.763
Gully-ERFNet	87.54	76.24	81.50	68.80	0.826

Table 4. Comparison of the proposed method with and without DEM data using the test set.

Model	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	Kappa
Gully-ERFNet	87.54	76.24	81.50	68.80	0.826
Gully-ERFNet + DEM	86.81	76.61	81.39	68.62	0.821

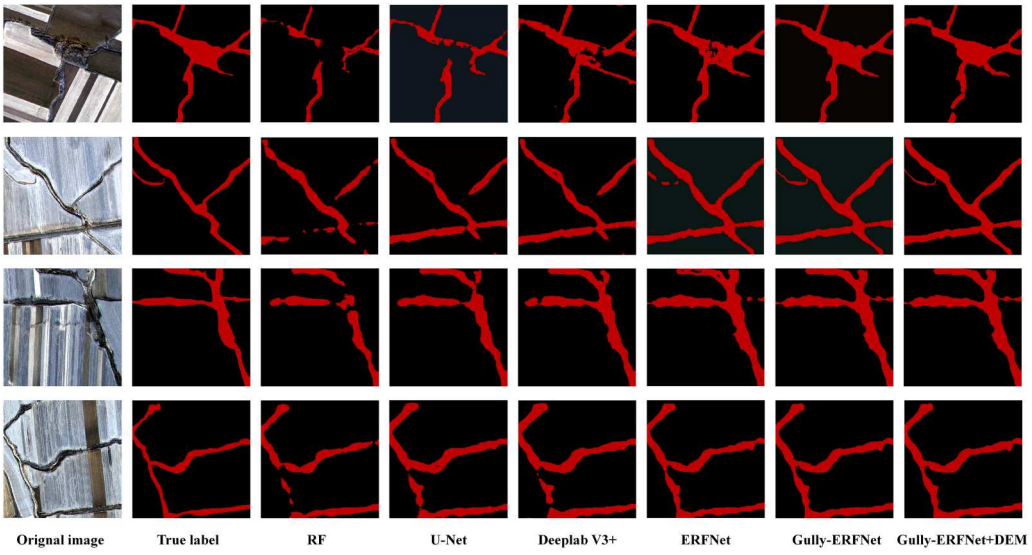

Figure 6. Comparison of model results for gully extraction: red indicates gullies, and black indicates non-gullies.

closely with the original markings and was particularly accurate in identifying smaller gullies. However, integrating the DEM data into the Gully-ERFNet model did not significantly enhance the continuity or accuracy, as evidenced by the discontinuities observed in the first and second test instances. These observations suggest that while DEM integration has potential, it may not always improve model performance. These results demonstrate the performance of various models in gully extraction tasks and indirectly validate the extraction precision of each model, as presented in Tables 3 and 4. It is worth noting that in the second image, a small gully near the main channel was overlooked by most models. Additionally, there is a horizontal road in this image that resembles gully features, all of which were mistakenly identified as gullies. This misidentification reduces the accuracy of the models. To mitigate interference from non-gully features such as roads, post-processing was applied to the extracted gullies, as described in Section 2.3.4. We selected a representative area within the study region to illustrate the effectiveness of this approach and compared the results before and after post-processing, as shown in Figure 7.

As shown in Figure 7, the post-processing method effectively removed small speckles and misclassified roads. However, the areas marked by yellow boxes indicate that owing to discontinuities in the model extraction process, some minor but actual gullies were inadvertently removed during

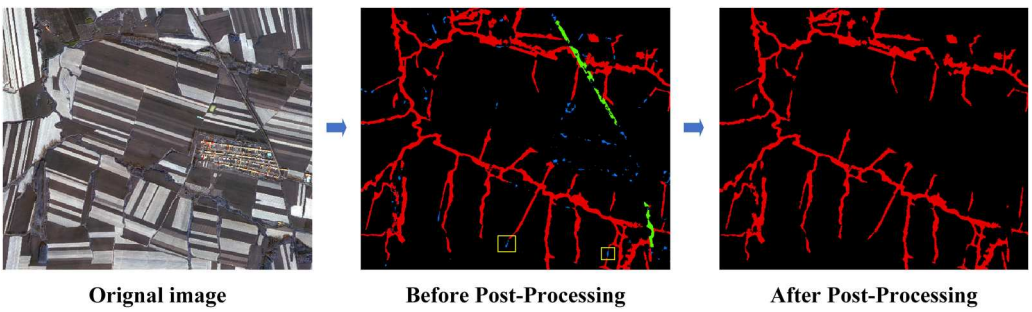

Figure 7. Comparison of gully extraction results before and after post-processing: red indicates gullies, black non-gully areas, blue removed spots, and green eliminated roads.

the elimination of small speckles. To further evaluate the effectiveness of the post-processing method, we compared the model performance metrics before and after post-processing within the study area, and the detailed results are presented in Table 5.

As shown in the Table 5, the application of the postprocessing method led to improvements across all evaluation metrics. The precision increased from 85.47% to 87.54%, and the recall improved from 73.14% to 76.24%, indicating a significant reduction in false positives and enhanced detection of actual erosion gullies. The F1-score rose to 81.50%, reflecting overall performance enhancement. Additionally, the IoU increased from 65.05% to 68.80%, and the Kappa coefficient improved from 0.794 to 0.826, further confirming the improvement in classification consistency. These results suggest that during the post-processing step of removing speckles smaller than 1000 pixels, the reduction in false positives contributed more to the performance improvement than the negative impact caused by the inadvertent removal of minor erosion gullies.

3.3. ACL function comparison experiment

To assess the influence of the ACL function on the Gully-ERFNet model for gully extraction, we analyzed the loss and accuracy curves for models trained with both the ACL and standard cross-entropy loss functions. This method provides a clear comparison of the performance differences between the two loss functions throughout the training process. In Figure 8, the blue line represents the loss curve obtained using the ACL function, whereas the red line illustrates the curve obtained without it. It is evident that the model employing the ACL function converges more rapidly, resulting in a significantly lower final loss value compared with the model without the ACL function. In terms of accuracy, the model with the ACL function attained a higher accuracy rate swiftly and stabilized sooner. Both curves indicate that using the ACL function leads to superior outcomes in the gully extraction tasks, highlighting its enhanced performance in gully data processing. These results underscore the effectiveness of the ACL function in mitigating the inaccuracy, noise, and class imbalance in gully data annotation by synergistically combining the focal and Dice loss functions.

Table 5. Comparison of the proposed method with and without DEM data using the test set.

Model	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	Kappa
Before Post-Processing	85.47	73.14	78.83	65.05	0.794
After Post-Processing	87.54	76.24	81.50	68.80	0.826

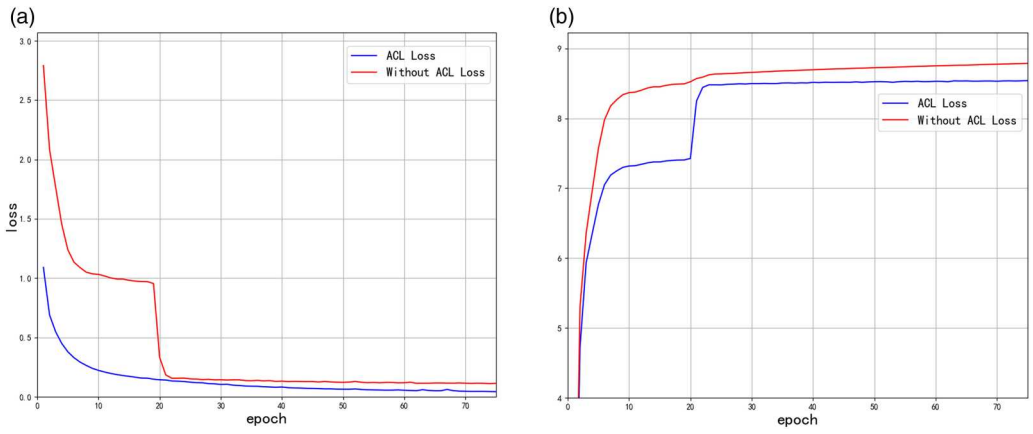

Figure 8. Training loss and accuracy curves of Gully-ERFNet with the ACL loss function and without the ACL loss function: (a) training loss curve; (b) accuracy curve.

3.4. Results of erosion gully extraction in the study area

In this study, Gully-ERFNet was employed to extract gullies from the two study areas, with two sample regions from each area presented, as illustrated in Figure 9. The gully extraction results for both study areas were relatively complete. In contrast, the results from the Hailun Basin demonstrated higher accuracy and greater complexity in the gully structures, with densely distributed gullies, intricate branching, and clear networks that cover almost all potential erosion areas. The extraction results for the Zake-Keyin Basin were more straightforward and concise and primarily depicted the main gully trunk and primary branches. Although the gully extraction results in

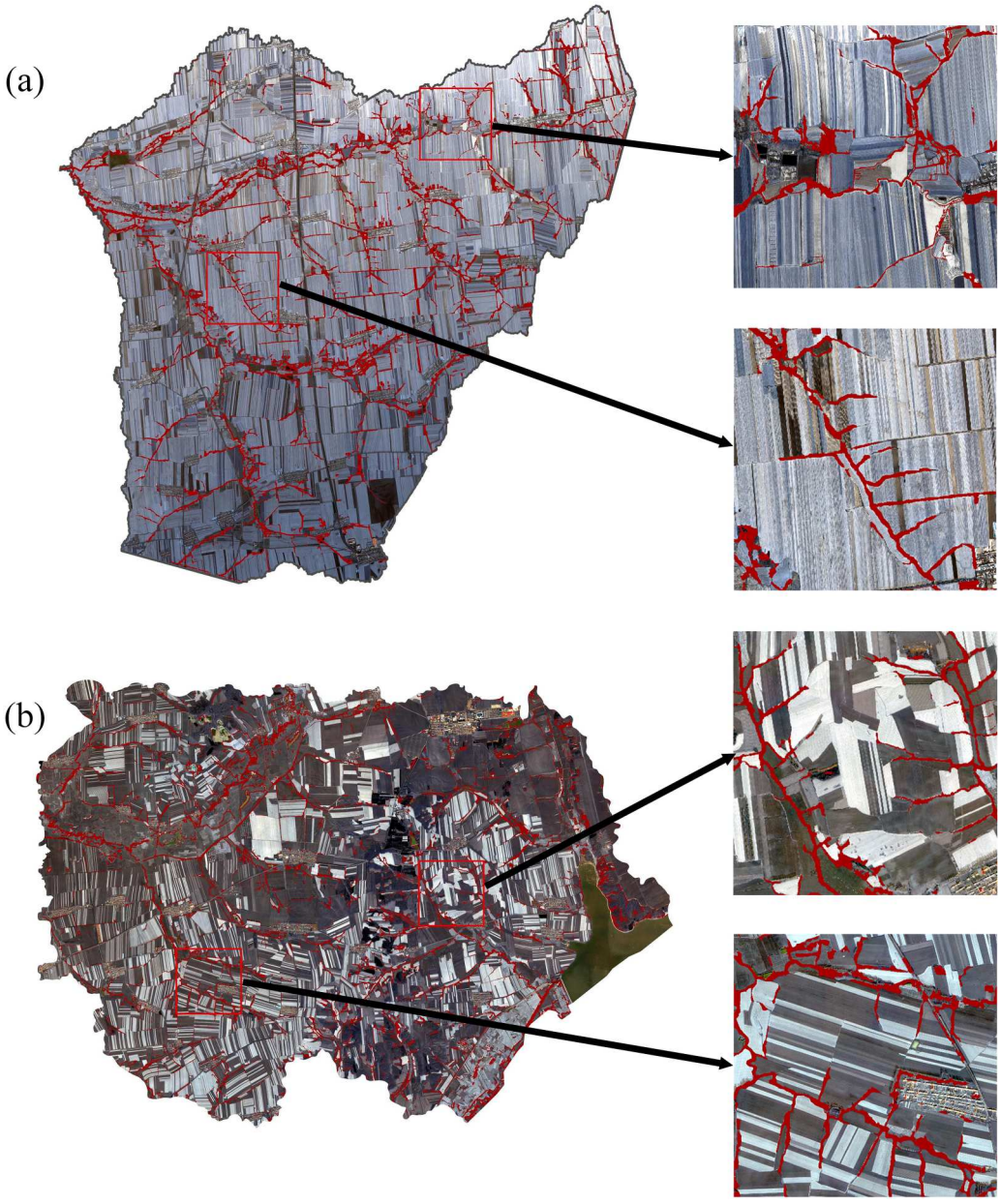

Figure 9. The spatial distribution of erosion gullies in the two study areas. (a) the erosion gully extraction results for the Hailun Basin; (b) the extraction results for the Zake-Keyin Basin.

Table 6. Comparison of evaluation metrics for test areas.

Model	Precision (%)	Recall (%)	F1-score (%)	IoU (%)	Kappa
Zake-Keyin	86.03	75.14	80.22	66.97	0.814
Hailun	89.05	77.34	82.78	70.62	0.842

this area were clear and easy to identify, they lacked detail, potentially omitting smaller gullies and finer structures.

To validate the adaptability and stability of the Gully-ERFNet model under different geographic conditions, this study utilized visual interpretation results from both regions as a reference standard for evaluating the outputs of the Gully-ERFNet model. The precision, recall, F1-score, IoU, and Kappa coefficient were calculated for each area, and the comparative results are presented in Table 6.

The results in the table demonstrate that the Gully-ERFNet model achieved higher performance in the Hailun Basin across all evaluation metrics. Specifically, the model attained a precision of 89.05%, recall of 77.34%, F1-score of 82.78%, IoU of 70.62%, and a Kappa coefficient of 0.842. In comparison, the Zake-Keyin Basin showed slightly lower values, with a precision of 86.03%, recall of 75.14%, F1-score of 80.22%, IoU of 66.97%, and a Kappa coefficient of 0.814. These results suggest that the model extracted gullies more accurately in the Hailun Basin, likely capturing gully structures more comprehensively and missing fewer small gullies. Conversely, the slightly lower recall and IoU in the Zake-Keyin Basin indicate potential limitations in detecting finer details or smaller gully features. Nonetheless, the high overall accuracy in both basins underscores the model's robustness and applicability for large-scale gully extraction in diverse terrains.

4. Discussion

4.1. Application of the Gully-ERFNet model

In this study, the Gully-ERFNet model was applied to extract erosion gullies under varying terrain conditions in the Hailun and Zake-Keyin Basins. The evaluation metrics in Table 4 and the spatial distribution maps in Figure 8 confirm the model's high feasibility and accuracy for gully extraction within the black soil region. The model effectively captures the complex details of gully networks in major channels and their tributaries across multiple scales. The precision and recall recorded in Table 4 were 89.05% and 77.34% for the Hailun Basin and 86.03% and 75.14% for the Zake-Keyin Basin, respectively, further demonstrating the reliability of the model. Moreover, despite the differences in terrain conditions and GF-2 imagery acquisition times between the two basins, the model exhibited strong consistency, with Kappa coefficients of 0.842 for the Hailun Basin and 0.814 for the Zake-Keyin Basin, reflecting the model's adaptability and stability across diverse conditions. As shown in Table 5, the application of post-processing techniques led to improvements across all evaluation metrics, whereas the comparative images in Figure 7 visually highlight the enhanced extraction results following post-processing. Although the post-processing method may inadvertently eliminate some small but genuine erosion gullies when minor patches are removed, this effect is negligible compared to the overall performance gains. These findings demonstrate that the Gully-ERFNet model effectively addresses the variations in both terrain and imagery. The incorporation of post-processing further enhances the extraction precision, ensures stable performance in gully detection, and reinforces the applicability and robustness of the model across diverse geographical environments (Adegun, Viriri, and Tapamo 2023; Zhu et al. 2024).

4.2. Analysis of the results of the comparison of different models

This study demonstrates the effectiveness of the proposed Gully-ERFNet model through comparative experiments utilizing five accuracy metrics: accuracy, recall, IoU, F1-score, and Kappa

coefficient. These metrics were used to assess the performance of the four models for gully erosion extraction. The data indicated that all deep-learning models surpassed the random forest method in these metrics, indicating the superior performance of deep-learning models in gully erosion extraction tasks compared with traditional machine-learning approaches.

In extracting erosional gullies, complex features such as terrain, texture, and edges pose challenges for conventional methods. In contrast, deep-learning models automatically learn more abstract and sophisticated feature representations from raw data. Additionally, these models generally possess larger parameter scales and higher complexities, offering greater flexibility and generalization capabilities than traditional machine-learning models (Ma et al. 2019). This makes them particularly adept at handling complex geospatial features such as erosion gullies and better suited for adapting to diverse data characteristics and scene variations. This supports our decision to employ deep learning models to improve the performance in this study (Chen et al. 2024).

Among the four deep-learning models, the Gully-ERFNet model achieved the best results for all evaluation metrics, and module ablation experiments showed that the introduction of the PPM module enhanced the capacity of the model to capture gully erosion features of various scales and shapes. Moreover, the SE attention mechanism helps the model focus better on important feature areas and improves its capability to capture details by dynamically adjusting the feature weights to different image contents. These improvements significantly enhanced the performance of the Gully-ERFNet model in gully erosion extraction tasks.

4.3. Impact of integrating DEM data on gully-ERFNet model performance

In this study, we integrated DEM data into the Gully-ERFNet model to assess its impact on the model performance. The findings indicate that incorporating DEM data did not significantly enhance the extraction accuracy of the model; in fact, a slight decrease was observed. This could be attributed to the relatively low spatial resolution of the DEM used in this study (12.5 m). Although bilinear interpolation was employed to resample the DEM to 0.8 meters to match the resolution of remote sensing imagery, this process struggles to accurately reconstruct true terrain features. Research has demonstrated that bilinear interpolation in DEM resampling leads to excessive smoothing of sharp terrain features such as gully edges and steep scarps, and results in the loss of high-frequency terrain information during the interpolation process (Minh et al. 2024; Shi, Wang, and Tian 2014; Walker, Wilkinson, and Hairsine 2024). Consequently, errors arising from insufficient resolution and noise interference during resampling may have led to blurred gully boundaries, introducing biases in the model's interpretation of topographic features and ultimately affecting its learning and predictive performance (Hawker et al. 2018; Huang, Su, et al. 2023; Lu et al. 2024). Moreover, even in the absence of DEM data, deep learning models can effectively identify gullies by leveraging image features such as texture, color, and shape, achieving high accuracy, as confirmed by (Gafurov and Yermolayev 2020; Huang et al. 2018; Wang et al. 2020) studies. Therefore, adding low-resolution DEM data to high-resolution imagery does not necessarily improve the accuracy of the deep-learning models.

However, it is important to note that these results primarily reflect the limitations of low-resolution DEM data and do not negate the potential value of DEM data for gully extraction. Utilizing high-resolution DEM data obtained from unmanned aerial vehicles (UAVs) or Light Detection and Ranging (LiDAR) technology can more accurately represent topographic details, providing a model with more valuable terrain information, and thereby improving gully extraction accuracy, as evidenced by the research of (Gevaert et al. 2018; Wang and Li 2021). However, acquiring high-resolution DEM data presents challenges, including high costs and limited coverage areas. For large-scale regional surveys, both time and financial expenditures increased significantly. Additionally, high-resolution DEM data require complex processing and calibration, which not only adds to the data preprocessing workload but also imposes higher demands on data processing workflows. In

practical applications, it is essential to balance data accuracy and acquisition costs to select the most appropriate DEM data source.

4.4. Limitations and future work

The Gully-ERFNet model demonstrated high accuracy in detecting erosion gullies in the black soil region of Northeast China using GF-2 remote sensing imagery. However, there remain areas for further improvement. First, the model's performance in identifying narrow gullies lacks sufficient stability, sometimes resulting in discontinuous detection outputs and reduced overall accuracy. Moreover, the study does not provide a comprehensive quantitative analysis of gullies with varying widths or those located in ambiguous regions, which limits a more nuanced understanding of the model's effectiveness. Second, although DEM data were incorporated into the study to evaluate their impact on model accuracy, related DEM-derived products (such as flow direction, slope, and curvature) have not undergone rigorous validation. This may constrain the model's ability to fully exploit topographic information for gully detection. Additionally, the study primarily relies on optical remote sensing data, leaving the potential of multi-source data fusion largely unexplored.

To overcome these limitations in detecting gully erosion within complex environments, future research should prioritize the synergistic integration of multi-source remote sensing data, particularly the deep fusion of thermal infrared (TIR) and hyperspectral data. TIR data can effectively capture subtle variations in surface temperature, which are closely linked to soil moisture content and vegetation cover – key driving factors influencing erosion processes (Khanal, Fulton, and Shearer 2017; Petropoulos et al. 2018). Incorporating TIR data into existing models could provide valuable spatio-temporal information on soil moisture dynamics, aiding in the identification of erosion-prone areas. Similarly, hyperspectral data, with its rich spectral resolution, offers detailed insights into soil mineral composition, organic matter content, and vegetation physiological status. These characteristics are critical for assessing soil erosion resistance and detecting early signs of erosion (Milewski et al. 2022; Wang et al. 2024). In addition, the systematic integration of DEM-derived products (such as flow direction, slope, and curvature) should also be strengthened, as these topographic parameters can help models more accurately predict runoff concentration areas and potential erosion paths (Chowdhuri et al. 2021). By establishing a deep fusion framework for multi-source data and fully utilizing the complementary advantages of various data types, future research is expected to significantly improve the ability of models to detect gully erosion at different scales and under various environmental conditions.

5. Conclusions

This paper proposes a novel semantic segmentation model, Gully-ERFNet, for gully extraction from the black soil region of Northeast China. The model integrates the PPM and SE attention mechanism modules into the traditional ERFNet framework and introduces the ACL function. Comparative experiments conducted on GF-2 remote sensing images from the Hailun and Zake-Keyin basins demonstrated that the Gully-ERFNet exhibited strong applicability and robustness under varying terrain conditions. Compared with the traditional ERFNet model, Gully-ERFNet achieved improvements of 3.85% in accuracy, 4.66% in F1 score, and 6.40% in IoU, with a Kappa coefficient of 0.826. These results indicate the reliability of the model in reducing misclassification and omission errors as well as its scalability and generalization capabilities in cross-basin extraction tasks. Additionally, post-processing methods effectively remove small patches and enhance the overall accuracy; although some small genuine erosion gullies may be inadvertently eliminated, the impact on the overall performance is negligible. Integration experiments with DEM data revealed that the inclusion of low-resolution DEM imagery does not significantly improve the classification accuracy, as relying solely on low-resolution topographic information is insufficient for capturing the detailed

features of erosion gullies. Furthermore, deep-learning methods outperform traditional machine-learning approaches in gully extraction, highlighting their advantages for terrain feature extraction. Ablation experiments further confirmed the positive impact of the PPM and SE attention modules on model performance. Loss-function comparison experiments indicate that the ACL function provides a more stable and effective solution for reducing loss and improving accuracy. Therefore, the Gully-ERFNet model has broad applicability and potential for widespread adoption, offering reliable technical support for gully monitoring and management in Northeast China. Future research could further explore multi-source data fusion and more refined topographic factors to comprehensively enhance the precision and robustness of gully erosion extraction.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This document is the result of a research project funded by the National Key RD Program of China [grant number 2021YFD1500102], Strategic Priority Research Program of the Chinese Academy of Sciences [grant number XDA28070502] and National Natural Science Foundation of China [grant number 42171380].

Data availability

Data will be made availability on request.

ORCID

Jiuchun Yang http://orcid.org/0000-0001-6211-0289

References

Adegun, Adekanmi Adeyinka, Serestina Viriri, and Jules-Raymond Tapamo. 2023. "Review of Deep Learning Methods for Remote Sensing Satellite Images Classification: Experimental Survey and Comparative Analysis." *Journal of Big Data* 10 (1): 93. https://doi.org/10.1186/s40537-023-00772-x.

Blaschke, Thomas, Geoffrey J Hay, Maggi Kelly, Stefan Lang, Peter Hofmann, Elisabeth Addink, Raul Queiroz Feitosa, Freek Van der Meer, Harald Van der Werff, and Frieke Van Coillie. 2014. "Geographic Object-Based Image Analysis – Towards a new Paradigm." *ISPRS Journal of Photogrammetry and Remote Sensing* 87:180–191. https://doi.org/10.1016/j.isprsjprs.2013.09.014.

Castillo, Carlos, Rafael Pérez, Michael R James, J. N. Quinton, Encarnación V. Taguas, and José A Gómez. 2012. "Comparing the Accuracy of Several Field Methods for Measuring Gully Erosion." *Soil Science Society of America Journal* 76 (4): 1319–1332. https://doi.org/10.2136/sssaj2011.0390.

Chen, Liangfu, Husi Letu, Meng Fan, Huazhe Shang, Jinhua Tao, Laixiong Wu, Ying Zhang, Chao Yu, Jianbin Gu, and Ning Zhang. 2022. "An Introduction to the Chinese High-Resolution Earth Observation System: Gaofen-1∼7 Civilian Satellites." *Journal of Remote Sensing*. https://doi.org/10.34133/2022/9769536.

Chen, Rong, Yi Zhou, Zetao Wang, Ying Li, Fan Li, and Feng Yang. 2024. "Towards Accurate Mapping of Loess Waterworn Gully by Integrating Google Earth Imagery and DEM Using Deep Learning." *International Soil and Water Conservation Research* 12 (1): 13–28. https://doi.org/10.1016/j.iswcr.2023.06.006.

Chen, Liang-Chieh, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. 2018. "Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation." Paper presented at the Proceedings of the European Conference on Computer Vision (ECCV).

Chowdhuri, Indrajit, Subodh Chandra Pal, Asish Saha, Rabin Chakrabortty, and Paramita Roy. 2021. "Evaluation of Different DEMs for Gully Erosion Susceptibility Mapping Using in-Situ Field Measurement and Validation." *Ecological Informatics* 65:101425. https://doi.org/10.1016/j.ecoinf.2021.101425.

Cong, Guojing, and Luca Buratti. 2018. "On Adam Trained Models and a Parallel Method to Improve the Generalization Performance." In *2018 IEEE/ACM Machine Learning in HPC Environments (MLHPC)*, 85–94. https://doi.org/10.1109/MLHPC.2018.8638641

Dagar, Jagdish Chander, and Anil Kumar Singh. 2018. *Ravine Lands: Greening for Livelihood and Environmental Security*. Vol. 636. Singapore: Springer.

De Santisteban, L. M., J. Casalí, and J. J. López. 2005. "Assessing Soil Erosion Rates in Cultivated Areas of Navarre (Spain)." *Earth Surface Processes and Landforms* 31 (4): 487–506. https://doi.org/10.1002/esp.1281.

Estornell, Javier, Jesus M. Martí-Gavliá, M. Teresa Sebastiá, and Jesus Mengual. 2013. "Principal Component Analysis Applied to Remote Sensing." *Modelling in Science Education and Learning* 6:83–89. https://doi.org/10.4995/msel.2013.1905.

Fathololoumi, Solmaz, Hiteshkumar B. Vasava, Daniel Saurette, Prasad Daggupati, and Asim Biswas. 2025. "Enhanced Ephemeral Gully Mapping Through Multi-Classifier Integration and Spectral Feature Analysis." *Catena* 248:108590. https://doi.org/10.1016/j.catena.2024.108590.

Gafurov, Artur M., and Oleg P. Yermolayev. 2020. "Automatic Gully Detection: Neural Networks and Computer Vision." *Remote Sensing* 12 (11): 1743. https://doi.org/10.3390/rs12111743.

Gao, Jay. 2009. "Bathymetric Mapping by Means of Remote Sensing: Methods, Accuracy and Limitations." *Progress in Physical Geography: Earth and Environment* 33 (1): 103–116. https://doi.org/10.1177/0309133309105657.

Gao, Yuhao, Yifan Zhu, Junxi Chen, Xue Yang, Yiting Huang, Fenghao Song, Yangbo He, et al. 2024. "Temporal and Spatial Distribution and Development of Permanent Gully in Cropland in the Rolling Hill Region (Phaeozems Area) of Northeast China." *Catena* 235:107625. https://doi.org/10.1016/j.catena.2023.107625.

Garosi, Younes, Mohsen Sheklabadi, Hamid Reza Pourghasemi, Ali Asghar Besalatpour, Christian Conoscenti, and Kristof Van Oost. 2018. "Comparison of Differences in Resolution and Sources of Controlling Factors for Gully Erosion Susceptibility Mapping." *Geoderma* 330:65–78. https://doi.org/10.1016/j.geoderma.2018.05.027.

Gevaert, C. M., C. Persello, F. Nex, and G. Vosselman. 2018. "A Deep Learning Approach to DTM Extraction from Imagery Using Rule-Based Training Labels." *ISPRS Journal of Photogrammetry and Remote Sensing* 142:106–123. https://doi.org/10.1016/j.isprsjprs.2018.06.001.

Ghassemian, Hassan. 2016. "A Review of Remote Sensing Image Fusion Methods." *Information Fusion* 32:75–89. https://doi.org/10.1016/j.inffus.2016.03.003.

Grossberg, Stephen. 2013. "Recurrent Neural Networks." *Scholarpedia* 8 (2): 1888. https://doi.org/10.4249/scholarpedia.1888.

Gu, Jiuxiang, Zhenhua Wang, Jason Kuen, Lianyang Ma, Amir Shahroudy, Bing Shuai, Ting Liu, et al. 2018. "Recent Advances in Convolutional Neural Networks." *Pattern Recognition* 77:354–377. https://doi.org/10.1016/j.patcog.2017.10.013.

Hawker, Laurence, Paul Bates, Jeffrey Neal, and Jonathan Rougier. 2018. "Perspectives on Digital Elevation Model (DEM) Simulation for Flood Modeling in the Absence of a High-Accuracy Open Access Global DEM." *Frontiers in Earth Science* 6:233. https://doi.org/10.3389/feart.2018.00233.

Hossain, Mohammad D., and Dongmei Chen. 2019. "Segmentation for Object-Based Image Analysis (OBIA): A Review of Algorithms and Challenges from Remote Sensing Perspective." *ISPRS Journal of Photogrammetry and Remote Sensing* 150:115–134. https://doi.org/10.1016/j.isprsjprs.2019.02.009.

Hu, Jie, Li Shen, and Gang Sun. 2018. "Squeeze-and-Excitation Networks." Paper presented at the Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.

Huang, Lingcao, Lin Liu, Liming Jiang, and Tingjun Zhang. 2018. "Automatic Mapping of Thermokarst Landforms from Remote Sensing Images Using Deep Learning: A Case Study in the Northeastern Tibetan Plateau." *Remote Sensing* 10 (12): 2067. https://doi.org/10.3390/rs10122067.

Huang, Donghao, Lin Su, Lili Zhou, Yulu Tian, and Haoming Fan. 2023. "Assessment of Gully Erosion Susceptibility Using Different DEM-Derived Topographic Factors in the Black Soil Region of Northeast China." *International Soil and Water Conservation Research* 11 (1): 97–111. https://doi.org/10.1016/j.iswcr.2022.04.001.

Huang, Xiaohui, Liyang Xiong, Yinghui Jiang, Sijin Li, Kai Liu, Hu Ding, and Guoan Tang. 2023. "Mapping Gully Affected Areas by Using Sentinel 2 Imagery and Digital Elevation Model Based on the Google Earth Engine." *Catena* 233:107473. https://doi.org/10.1016/j.catena.2023.107473.

Jadon, Shruti. 2020. "A Survey of Loss Functions for Semantic Segmentation." Paper presented at the 2020 IEEE conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB).

Karami, A., A. Khoorani, A. Noohegar, S. R. F. Shamsi, and V. Moosavi. 2015. "Gully Erosion Mapping Using Object-Based and Pixel-Based Image Classification Methods." *Environmental & Engineering Geoscience* 21 (2): 101–110. https://doi.org/10.2113/gseegeosci.21.2.101.

Khanal, Sami, John Fulton, and Scott Shearer. 2017. "An Overview of Current and Potential Applications of Thermal Remote Sensing in Precision Agriculture." *Computers and Electronics in Agriculture* 139:22–32. https://doi.org/10.1016/j.compag.2017.05.001.

Kotaridis, Ioannis, and Maria Lazaridou. 2021. "Remote Sensing Image Segmentation Advances: A Meta-Analysis." *ISPRS Journal of Photogrammetry and Remote Sensing* 173:309–322. https://doi.org/10.1016/j.isprsjprs.2021.01.020.

Li, Bingyao, Jingming Hou, Donglai Li, Dong Yang, Hao Han, Xu Bi, Xinghua Wang, Reinhard Hinkelmann, and Junqiang Xia. 2021. "Application of LiDAR UAV for High-Resolution Flood Modelling." *Water Resources Management* 35 (5): 1433–1447. https://doi.org/10.1007/s11269-021-02783-w.

Li, Deren, Mi Wang, and Jie Jiang. 2020. "China's High-Resolution Optical Remote Sensing Satellites and Their Mapping Applications." *Geo-spatial information science* 24 (1): 85–94. https://doi.org/10.1080/10095020.2020.1838957.

Lin, Tsung-Yi, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. "Focal Loss for Dense Object Detection." Paper presented at the proceedings of the IEEE International Conference on Computer Vision.

Liu, Y., D. Wang, J. Gao, and W. Deng. 2005. "Land Use/Cover Changes, the Environment and Water Resources in Northeast China." *Environmental Management* 36 (5): 691–701. https://doi.org/10.1007/s00267-004-0285-5.

Liu, Boyang, Biao Zhang, Hao Feng, Shufang Wu, Jiangtao Yang, Yufeng Zou, and Kadambot H. M. Siddique. 2022. "Ephemeral Gully Recognition and Accuracy Evaluation Using Deep Learning in the Hilly and Gully Region of the Loess Plateau in China." *International Soil and Water Conservation Research* 10 (3): 371–381. https://doi.org/10.1016/j.iswcr.2021.10.004.

Liu, Honghu, Tianyu Zhang, Baoyuan Liu, Gang Liu, and G. V. Wilson. 2012. "Effects of Gully Erosion and Gully Filling on Soil Depth and Crop Production in the Black Soil Region, Northeast China." *Environmental Earth Sciences* 68 (6): 1723–1732. https://doi.org/10.1007/s12665-012-1863-0.

Lu, Pingda, Bin Zhang, Chenfeng Wang, Mengyun Liu, and Xiaoping Wang. 2024. "Erosion Gully Networks Extraction Based on InSAR Refined Digital Elevation Model and Relative Elevation Algorithm—A Case Study in Huangfuchuan Basin, Northern Loess Plateau, China." *Remote Sensing* 16 (5): 921. https://doi.org/10.3390/rs16050921.

Ma, Lei, Yu Liu, Xueliang Zhang, Yuanxin Ye, Gaofei Yin, and Brian Alan Johnson. 2019. "Deep Learning in Remote Sensing Applications: A Meta-Analysis and Review." *ISPRS Journal of Photogrammetry and Remote Sensing* 152:166–177. https://doi.org/10.1016/j.isprsjprs.2019.04.015.

Ma, Rui, Zhiyuan Tian, Yan Zhao, Yihang Wu, and Yin Liang. 2024. "Response of Soil Quality Degradation to Cultivation and Soil Erosion: A Case Study in a Mollisol Region of Northeast China." *Soil and Tillage Research* 242:106159. https://doi.org/10.1016/j.still.2024.106159.

Maćkiewicz, Andrzej, and Waldemar Ratajczak. 1993. "Principal Components Analysis (PCA)." *Computers & Geosciences* 19 (3): 303–342. https://doi.org/10.1016/0098-3004(93)90090-r.

Milewski, Robert, Thomas Schmid, Sabine Chabrillat, Marcos Jiménez, Paula Escribano, Marta Pelayo, and Eyal Ben-Dor. 2022. "Analyses of the Impact of Soil Conditions and Soil Degradation on Vegetation Vitality and Crop Productivity Based on Airborne Hyperspectral VNIR–SWIR–TIR Data in a Semi-Arid Rainfed Agricultural Area (Camarena, Central Spain)." *Remote Sensing* 14 (20): 5131. https://doi.org/10.3390/rs14205131.

Minh, Nguyen Quang, Nguyen Thi Thu Huong, Pham Quoc Khanh, La Phu Hien, and Dieu Tien Bui. 2024. "Impacts of Resampling and Downscaling Digital Elevation Model and its Morphometric Factors: A Comparison of Hopfield Neural Network, Bilinear, Bicubic, and Kriging Interpolations." *Remote Sensing* 16 (5): 819. https://doi.org/10.3390/rs16050819.

Mollick, Taposh, Md Golam Azam, and Sabrina Karim. 2023. "Geospatial-based Machine Learning Techniques for Land Use and Land Cover Mapping Using a High-Resolution Unmanned Aerial Vehicle Image." *Remote Sensing Applications: Society and Environment* 29:100859. https://doi.org/10.1016/j.rsase.2022.100859.

Neupane, Bipul, Teerayut Horanont, and Jagannath Aryal. 2021. "Deep Learning-Based Semantic Segmentation of Urban Features in Satellite Images: A Review and Meta-Analysis." *Remote Sensing* 13 (4): 808. https://doi.org/10.3390/rs13040808.

Petropoulos, G. P., P. K. Srivastava, K. P. Ferentinos, and D. Hristopoulos. 2018. "Evaluating the Capabilities of Optical/TIR Imaging Sensing Systems for Quantifying Soil Water Content." *Geocarto International* 35 (5): 494–511. https://doi.org/10.1080/10106049.2018.1520926.

Phinzi, Kwanele, Imre Holb, and Szilárd Szabó. 2021. "Mapping Permanent Gullies in an Agricultural Area Using Satellite Images: Efficacy of Machine Learning Algorithms." *Agronomy* 11 (2): 333. https://doi.org/10.3390/agronomy11020333.

Rigatti, S. J. 2017. "Random Forest." *Journal of Insurance Medicine* 47 (1): 31–39. https://doi.org/10.17849/insm-47-01-31-39.1.

Romera, Eduardo, Jose M. Alvarez, Luis M. Bergasa, and Roberto Arroyo. 2018. "ERFNet: Efficient Residual Factorized ConvNet for Real-Time Semantic Segmentation." *IEEE Transactions on Intelligent Transportation Systems* 19 (1): 263–272. https://doi.org/10.1109/tits.2017.2750080.

Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. 2015. "U-net: Convolutional Networks for Biomedical Image Segmentation." Paper presented at the Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18.

Shahabi, H., B. Jarihani, S. Tavakkoli Piralilou, D. Chittleborough, M. Avand, and O. Ghorbanzadeh. 2019. "A Semi-Automated Object-Based Gully Networks Detection Using Different Machine Learning Models: A Case Study of Bowen Catchment, Queensland, Australia." *Sensors* 19 (22): 4893. https://doi.org/10.3390/s19224893.

Shen, Yi, Nan Su, Chunhui Zhao, Yiming Yan, Shou Feng, Yong Liu, and Wei Xiang. 2024. "A Foreground-Driven Fusion Network for Gully Erosion Extraction Utilizing UAV Orthoimages and Digital Surface Models." *IEEE Transactions on Geoscience and Remote Sensing* 62:1–16. https://doi.org/10.1109/tgrs.2024.3417398.

Shi, Wenzhong, Bin Wang, and Yan Tian. 2014. "Accuracy Analysis of Digital Elevation Model Relating to Spatial Resolution and Terrain Slope by Bilinear Interpolation." *Mathematical Geosciences* 46 (4): 445–481. https://doi.org/10.1007/s11004-013-9508-8.

Shivhare, Priyanka, and Vinay Gupta. 2015. "Review of Image Segmentation Techniques Including Pre & Post Processing Operations." *International Journal of Engineering and Advanced Technology* 4 (3): 153–157.

Sidorchuk, A. 1999. "Dynamic and Static Models of Gully Erosion." *Catena* 37 (3-4): 401–414. https://doi.org/10.1016/s0341-8162(99)00029-6.

Singh, Harbhajan, and Vijay Dhir. 2019. "Effectiveness of Python Libraries in Machine Learning: A Review." *Webology (ISSN: 1735-188X)* 16 (1).

Tang, Jie, Gang Liu, Yun Xie, Xingwu Duan, Daan Wang, and Shuai Zhang. 2021. "Ephemeral Gullies Caused by Snowmelt: A Ten-Year Study in Northeastern China." *Soil and Tillage Research* 212:105048. https://doi.org/10.1016/j.still.2021.105048.

Tang, Jie, Yun Xie, Chuan Liu, Hanlin Dong, and Gang Liu. 2022. "Effects of Rainfall Characteristics and Contour Tillage on Ephemeral Gully Development in a Field in Northeastern China." *Soil and Tillage Research* 218:105312. https://doi.org/10.1016/j.still.2021.105312.

Tang, Jie, Yun Xie, Yongqiu Wu, and Gang Liu. 2023. "Influence of Precipitation Change and Topography Characteristics on the Development of Farmland Gully in the Black Soil Region of Northeast China." *Catena* 224:106999. https://doi.org/10.1016/j.catena.2023.106999.

Thwaites, Robin N., Andrew P. Brooks, Timothy J. Pietsch, and John R. Spencer. 2021. "What Type of Gully is That? The Need for a Classification of Gullies." *Earth Surface Processes and Landforms* 47 (1): 109–128. https://doi.org/10.1002/esp.5291.

Walker, Simon J., Scott N. Wilkinson, and Peter B. Hairsine. 2024. "Advancing Gully Topographic Threshold Analysis Using an Automated Algorithm and High-Resolution Topography." *Catena* 239:107897. https://doi.org/10.1016/j.catena.2024.107897.

Wang, Sizhe, and Wenwen Li. 2021. "GeoAI in Terrain Analysis: Enabling Multi-Source Deep Learning and Data Fusion for Natural Feature Detection." *Computers, Environment and Urban Systems* 90:101715. https://doi.org/10.1016/j.compenvurbsys.2021.101715.

Wang, Zhaobin, Yikun Ma, and Yaonan Zhang. 2023. "Review of Pixel-Level Remote Sensing Image Fusion Based on Deep Learning." *Information Fusion* 90:36–58. https://doi.org/10.1016/j.inffus.2022.09.008.

Wang, Jiaqi, Jiuchun Yang, Zhi Li, Liwei Ke, Qingyao Li, Jianwei Fan, and Xue Wang. 2024. "Research on Soil Erosion Based on Remote Sensing Technology: A Review." *Agriculture* 15 (1): 18. https://doi.org/10.3390/agriculture15010018.

Wang, Wenting, Shuiqing Yin, Juan Yu, Zeng He, and Yun Xie. 2023. "Long-term Trends of Precipitation and Erosivity Over Northeast China During 1961–2020." *International Soil and Water Conservation Research* 11 (4): 743–754. https://doi.org/10.1016/j.iswcr.2023.04.002.

Wang, Ranghu, Shuwen Zhang, Luoman Pu, Jiuchun Yang, Chaobin Yang, Jing Chen, Cong Guan, et al. 2016. "Gully Erosion Mapping and Monitoring at Multiple Scales Based on Multi-Source Remote Sensing Data of the Sancha River Catchment, Northeast China." *ISPRS International Journal of Geo-Information* 5 (11): 200. https://doi.org/10.3390/ijgi5110200.

Wang, Biwei, Zengxiang Zhang, Xiao Wang, Xiaoli Zhao, Ling Yi, and Shunguang Hu. 2020. "Object-Based Mapping of Gullies Using Optical Images: A Case Study in the Black Soil Region, Northeast of China." *Remote Sensing* 12 (3): 487. https://doi.org/10.3390/rs12030487.

Wen, Yanru, Till Kasielke, Hao Li, Harald Zepp, and Bin Zhang. 2021. "A Case-Study on History and Rates of Gully Erosion in Northeast China." *Land Degradation & Development* 32 (15): 4254–4266. https://doi.org/10.1002/ldr.4031.

Xu, Hongzhang, Hongjie He, Ying Zhang, Lingfei Ma, and Jonathan Li. 2023. "A Comparative Study of Loss Functions for Road Segmentation in Remotely Sensed Road Datasets." *International Journal of Applied Earth Observation and Geoinformation* 116:103159. https://doi.org/10.1016/j.jag.2022.103159.

Xu, X. Z., Y. Xu, S. C. Chen, S. G. Xu, and H. W. Zhang. 2010. "Soil Loss and Conservation in the Black Soil Region of Northeast China: A Retrospective Study." *Environmental Science & Policy* 13 (8): 793–800. https://doi.org/10.1016/j.envsci.2010.07.004.

Yu, Bo, Lu Yang, and Fang Chen. 2018. "Semantic Segmentation for High Spatial Resolution Remote Sensing Images Based on Convolution Neural Network and Pyramid Pooling Module." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 11 (9): 3252–3261. https://doi.org/10.1109/jstars.2018.2860989.

Zeiler, Matthew D, and Rob Fergus. 2014. "Visualizing and Understanding Convolutional Networks." Paper presented at the computer vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part I 13.

Zhang, Shengmin, Xing Han, Richard M. Cruse, Xingyi Zhang, Wei Hu, Yue Yan, and Mingming Guo. 2022. "Morphological Characteristics and Influencing Factors of Permanent Gully and its Contribution to Regional Soil Loss Based on a Field Investigation of 393 km2 in Mollisols Region of Northeast China." *Catena* 217:106467. https://doi.org/10.1016/j.catena.2022.106467.

Zhang, Kuayue, Qingmin Liao, Juncheng Zhang, Shaojun Liu, Haoyu Ma, and Jing-Hao Xue. 2021. "EFRNet: A Lightweight Network with Efficient Feature Fusion and Refinement for Real-Time Semantic Segmentation." Paper presented at the 2021 IEEE International Conference on Multimedia and Expo (ICME).

Zhang, Xinle, Shengqi Zhang, Xiangtian Meng, Guowei Zhang, Deqiang Zang, Yongqi Han, Hongfu Ai, and Huanjun Liu. 2024. "Remote Sensing Image Segmentation of Gully Erosion in a Typical Black Soil Area in Northeast China Based on Improved DeepLabV3+ Model." *Ecological Informatics* 84:102929. https://doi.org/10.1016/j.ecoinf.2024.102929.

Zhao, Hengshuang, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. 2017. "Pyramid Scene Parsing Network." Paper presented at the proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.

Zhu, Panpan, Hao Xu, Ligang Zhou, Peixin Yu, Liqiang Zhang, and Suhong Liu. 2024. "Automatic Mapping of Gully from Satellite Images Using Asymmetric Non-Local LinkNet: A Case Study in Northeast China." *International Soil and Water Conservation Research* 12 (2): 365–378. https://doi.org/10.1016/j.iswcr.2023.07.006.